

Integer Parts, Fractional Parts, and the Division Algorithm

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Motivation

How do integers sit inside the real numbers? Every real number seems to lie "between" two consecutive integers. The Theorem makes this intuition precise by showing that every real number can be decomposed uniquely into an integer part and a fractional part. In other words, every real number admits a canonical representation of a whole-number component and a remainder of less than one. This result is fundamental because it provides a bridge between the continuous world of real numbers and the discrete world of integers. The integer q records the position of x among the integers, while the quantity s measures the distance from x to that integer. The uniqueness of this decomposition ensures that every real number has a well-defined integer part and a fractional part. Also, this result can be used to deduce the Division Algorithm which will show us the implication of this result to one of the most fundamental result of Number Theory

Theorem 1. *For every real number x , there exist a unique integer q and a unique real number s satisfying*

$$x = q + s, \quad 0 \leq s < 1.$$

The proof of the theorem is conducted in two stages. We first establish the existence of an integer q satisfying

$$q \leq x < q + 1$$

To obtain such an integer, we consider the set of all integers less than or equal to x and use the least upper bound property of the real numbers. Once such a q has been found, defining

$$s = x - q$$

Naturally, the decomposition $x = q + s$, and it remains only to verify that $0 \leq s < 1$. Finally, we show that no other pair (q, s) can satisfy these conditions, thus establishing the uniqueness.

Proof. Let x be a real number and consider the set A which includes all integers less than or equal to x . By the Archimedean Property, there exists an integer N such that $N > x$. Then $-N \leq x$, so $-N \in A$. Hence, A is nonempty. Since x is an upper bound for A , there exists a greatest integer q such that $q \leq x$. Because q is the greatest such integer, $q + 1 > x$. Hence,

$$q \leq x < q + 1$$

Subtracting q throughout gives $0 \leq x - q < 1$. Define $s = x - q$ which follows $x = q + s$

Now to prove uniqueness, assume that $x = q_1 + s_1$ where q_1 is an integer and s_1 is a real number such that $0 \leq s_1 < 1$. Then $q - q_1 = s_1 - s$ and $0 \leq s_1, s < 1$, so $-1 < s_1 - s < 1$. Hence, $-1 < q - q_1 < 1$ as we know both q and q_1 are integers constrained in the inequality gives us $q - q_1 = 0$ which follows $q = q_1$, consequently $s = s_1$. Hence Uniqueness is now established $\square$

Now this result can be used to deduce the Division Algorithm. Applying the above result to the real number $x = \frac{a}{b}$, where a and b are integers, so $\frac{a}{b} = q + s$, multiplying b throughout $a = bq + bs$, defining $bs = r$ hence, $0 \leq r < b$. That gives $a = bq + r$ which is precisely the Division Algorithm

Geometric Interpretation

The theorem implies a natural geometric interpretation of the real line. The integers partition the real line into half-open intervals of the form $[n, n + 1)$ where $n \in \mathbb{Z}$.

The theorem states that every real number belongs to exactly one such interval. Indeed, if

$$x = q + s, \quad 0 \leq s < 1$$

then

$$q \leq x < q + 1$$

Thus q identifies the unique unit interval containing x , while s measures the distance of x from the left endpoint q . In this sense, every real number may be viewed as consisting of an integer part together with its position inside the corresponding unit interval.

Concluding Remarks

The geometric interpretation of the theorem provides additional insight into the division algorithm. The theorem partitions the real line into unit intervals with the position of a real number within its interval. Similarly, the division algorithm partitions the integers into blocks of size b as follows :

$$[0, b), [b, 2b), [2b, 3b), \dots$$

The quotient identifies the block containing a given integer, while the remainder records its position within that block. In this sense, the division algorithm maybe viewed as the discrete analogue of the decomposition of a real number into its integer and fractional parts scaled to b .

The theorem appears elementary, but it provides a canonical decomposition of every real number into an integer part and a fractional part. Moreover, the division algorithm arises naturally from this decomposition, illustrating how a statement about arbitrary real numbers leads directly to a fundamental result of number theory which is the division algorithm.